Big Data and Machine Learning for Applied Economics

Week 01: Predict, Don't Explain — The Statistical Learning Framework

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About this course

1. **By the end of the course you will be able to:**
   - Predict and classify economic variables with ML methods and evaluate performance *out of sample*, honestly.
   - Build the full workflow in R: splitting, validation, fitting, evaluation, and interpretation.
   - Open up a flexible model (variable importance, partial dependence, SHAP) and know what it says and what it does not.
   - Read critically and present economic research that uses ML.
2. **Textbook:** James, Witten, Hastie & Tibshirani, *An Introduction to Statistical Learning* (ISL, 2nd ed.). We follow its progression.
3. **Methodology:** for each method, the problem it solves $\rightarrow$ intuition $\rightarrow$ essential formulation $\rightarrow$ how it works $\rightarrow$ application with real data $\rightarrow$ evaluation $\rightarrow$ interpretation. Long derivations go to each session's "Going deeper" appendix.
4. **Grading:** paper presentation (10%) - Problem Set 1 (15%) - Problem Set 2 (15%) - Final project (60%).

The map of the 9 sessions

#	Topic	ISL
1	Predict, don't explain: the statistical learning framework ← **today**	1–2
2	Linear regression as a prediction machine; evaluating a prediction	2–3
3	Cross-validation and regularization: the honest pipeline	5–6
4	Classification: from the logit to decisions	4
5	Trees, forests, and boosting	8
6	Opening the black box: importance, partial dependence, and SHAP	—
7	Representation: PCA, k-means, and neural networks	12, 10
8	The papers and the synthesis: student presentations (Nov 3)	—
9	Final project presentations (Nov 24)	—

- **Problem Set 1:** sessions 2–4 (out October 6, due the 20th). **Problem Set 2:** sessions 5–7 (October 20 to November 10). Applied only, on Cali housing, with a ranking by test error.
- In class, from session 2 through 8, a single problem: the vote at Colombia's polling stations. In session 8 each student presents a **paper** from their area (15 minutes); papers are chosen in session 2.
- The **final project** is a predictive research proposal: the idea on November 3, the document and presentation on the 24th.

The data that stay with us all semester

Three uses of real data, so the effort goes into the methods:

1. **In class: Colombia's polling stations.** The winner's vote share and turnout at each station, linked to the surrounding census (stratum, services, ages, ethnicity) and the location. One problem from session 2 through 8.
2. **In the problem sets: housing prices in Cali.** Tens of thousands of listings with attributes and location, to transfer the tools to another problem.
3. **In the final project: whatever dataset each student picks** for their own question.

The question that opens the course

How predictable is an election from the *stable* characteristics of places? We answer it in stages across the course: first with a regression (session 2), then with a forest (session 5), then by opening the forest (session 6).

R leveling and AI policy

R leveling (required, self-paced, outside class hours):

- Introduction to R: Available [here]
- Data manipulation with `dplyr`: Available [here]
- Reading and writing data: Available [here]
- Data management and organization: Available [here]
- Good modeling practice with `tidymodels`: material on **Intu**.
- For those taking the Python route, equivalent reference material is available.

AI policy

AI tools are allowed in any component of the course, as long as you **explicitly declare** what was used and for what.

1. From β to y: two questions for the same data
2. The statistical learning framework
3. Overfitting and complexity
4. Bias and variance: why the test error is U-shaped
5. The task map and the course workflow

The same data, two questions

Take the polling stations: the winner's vote share (y) and the ethnic composition of the surroundings (x).

Causal question (Econometrics)

- How much does the vote *change* if the ethnic composition changes?
- Object: a parameter β.
- Enemy: bias (selection, omitted variables).
- Validated with an identification **design**.

Predictive question (this course)

- How well can we *anticipate* the vote at a station we have never seen, and what does that depend on?
- Object: the whole function $\hat{f}(x)$ and its error.
- Enemy: overfitting (variance).
- Validated with **new data**.

The sentence we will repeat all semester

A significant coefficient is not a prediction, and a variable's importance in a predictive model is not a causal effect. They are different questions, with different tools and different success criteria.

So what is machine learning?

- A branch of statistics and computer science that develops **algorithms to predict** an outcome y from observable variables x.
- The "learning" part: **we do not specify** in advance the exact way the machine should predict y from x; that is left as an empirical problem the algorithm solves with the data.
- The approach is pragmatic: we abstract away from the underlying model.

"Whatever works, works. . . "

- **How is it different from the econometrics you already know?**
  - Classical econometrics: estimate **parameters** (β) with desirable properties and interpret them.
  - ML: produce **predictions** ($\hat{y}$) that work well on **new** data, and then understand what they depend on.

ML's first great victory: Google Flu Trends

Context (2008): in the U.S., flu surveillance depended on reports that medical centers send to the CDC.

- The CDC aggregates the reports by region and nationally.
- The whole process took $\approx$ **10 days** $\rightarrow$ too long for an epidemic.

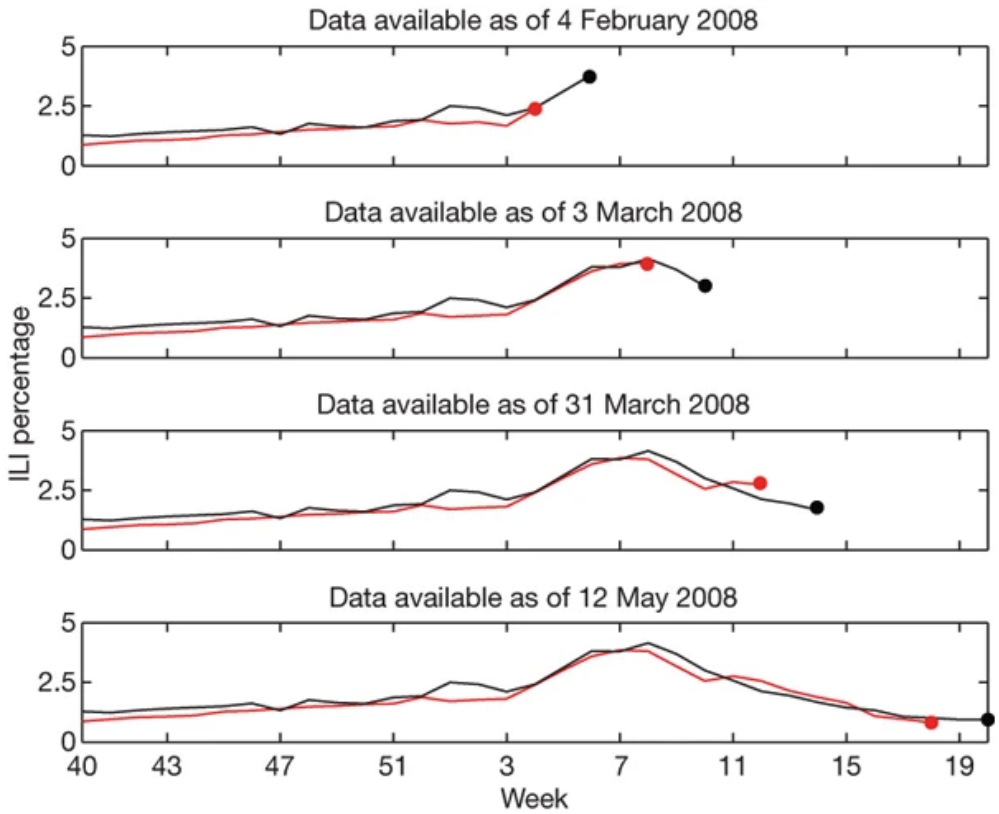
Google's proposal (Ginsberg et al., 2009, *Nature*):

- Match the official data with flu-related **searches**.
- They evaluated $\approx$ **50 million** candidate terms and $\approx$ **450 million** models; the final model used 45 terms.
- Result: near **real-time** flu-intensity predictions, for any region.
- What took the CDC **10 days** took Google **1**.

No epidemiological theory, no surveys, no structural model. Convinced?

The figure that convinced the world

Ginsberg et al. (2009, *Nature*), Fig. 3: what was known at four dates in 2007–08



U.S. Mid-Atlantic region. Black: % of medical visits for influenza-like illness (ILI) estimated by Google's model from that same week's searches. Red: the official CDC figure, which arrives with a one-to-two-week lag. Source: Ginsberg et al. (2009), Fig. 3.

A curve built from Google searches tracks the official curve week by week, and arrives earlier. What could go wrong?

. . . and its first great defeat

"The Parable of Google Flu" (Lazer et al., 2014, *Science*)

- In the 2012–2013 season, Google Flu Trends predicted **more than double** the cases that actually occurred.
- In fact, it overestimated prevalence in **100 of the 108 weeks** after August 2011.
- **What went wrong?**
  - With 50 million terms, many correlate with the flu **by chance** (e.g., seasonal basketball searches).
  - Search behavior **changes over time** (and Google changes its search engine): the model was trained on a world that stopped existing.
  - "*Big data hubris*": the implicit idea that data size substitutes for careful measurement and validation.

The lesson that organizes this course

Predicting well **within** the sample is easy; the hard part is **generalizing** to new data. Today's entire apparatus (risk, overfitting, bias–variance) exists to attack that problem.

Umbrella or rain dance?

Not every policy question needs a causal effect. Compare:

- **Should I do the rain dance?** It only matters whether the dance *causes* rain $\rightarrow$ a **causal** problem (β).
- **Should I carry an umbrella?** It only matters whether it will rain, not why $\rightarrow$ a **prediction** problem ($\hat{y}$).

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Real examples of prediction problems in public policy:

- **Cadastral appraisal:** the property tax is levied on the value of every parcel in the city, and almost none were sold this year. It does not matter *why* it is worth what it is worth; what matters is *how much*, and that has to be predicted.
- **Pretrial release** (Kleinberg et al., 2018, QJE): judges decide on an implicit prediction of flight or reoffending. An ML model could cut crime by $\approx 25\%$ with no more inmates, or the jail population by $\approx 42\%$ with no more crime.
- **Targeting:** Colombia's Sisbén is, in essence, a household **classifier** — it predicts structural poverty from observable characteristics.

Is my question a β question or a $\hat{y}$ question?

A practical rule: what will I do with the answer?

- β **problem:** the decision is to **change** x; I need the effect on y and a design that identifies it.
- $\hat{y}$ **problem:** the decision only needs to **know** what value y will take; I need a good predictor and an honest test.

β **questions** (the decision changes x)

- Does a home's price rise if a metro station is built next door?
- Does the scholarship improve Saber Pro scores?
- Does an area's vote change if service coverage improves?

$\hat{y}$ **questions** (knowing what will happen is enough)

- How much is this home worth?
- Which students will drop out this semester?
- At which stations will the candidate win?

The two questions are not rivals

A model can predict very well with meaningless coefficients, and a credible β can predict badly. This course optimizes $\hat{y}$; at the end we will see how a good $\hat{y}$ helps estimate β.

1 From β to y: two questions for the same data

2 The statistical learning framework

3 Overfitting and complexity

4 Bias and variance: why the test error is U-shaped

5 The task map and the course workflow

The setup

Data: we observe n pairs (x_i, y_i), drawn i.i.d. from an **unknown** joint distribution $P(X, Y)$.

- Y: the response variable (wage, default, consumption, price, vote).
- $X = (X_1, \ldots, X_p)$: predictors or features.

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Fundamental decomposition: we can always write

$$Y \;=\; f(X) \;+\; \varepsilon, \qquad \text{con } f(x) \equiv \mathbb{E}[Y \mid X = x]$$

- f: the **regression function** — the systematic part of Y.
- $\varepsilon \equiv Y - f(X)$: what X does not explain. **By construction** $\mathbb{E}[\varepsilon \mid X] = 0$, with $\mathrm{Var}(\varepsilon) = \sigma^2$.

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The goal of supervised learning

Use the data to build $\hat{f} \approx f$. In econometrics we cared about a **parameter inside** f; here we care about the **whole function**, because that is what we predict with.

What exactly does "learning" mean?

Learning = choosing $\hat{f}$ within a **class of functions** $\mathcal{F}$ using the data:

- $\mathcal{F}$ can be: lines, polynomials, trees, neural networks...
- Each session of this course is, at bottom, a different class $\mathcal{F}$.

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And what is a "good" $\hat{f}$? We need a **loss function** $L(y, \hat{f}(x))$ that measures the cost of being wrong:

- **Squared:** $L = (y - \hat{f}(x))^2$. Punishes large errors hard; the standard in regression.
- **Absolute:** $L = |y - \hat{f}(x)|$. More robust to extreme values.
- **0–1:** $L = \mathbb{I}\{y \neq \hat{f}(x)\}$. For classification (session 4).

Choosing L is an economic decision

Which error costs more? Underestimating demand (lost sales) is not the same as overestimating it (dead inventory); nor a false positive the same as a false negative when targeting a subsidy.

The generalization risk

Definition. The **risk** (or generalization error) of a predictor f is its expected loss on a **new** data point drawn from P:

$$R(f) \;=\; \mathbb{E}\big[L(Y, f(X))\big]$$

With squared loss, $R(f) = \mathbb{E}[(Y - f(X))^2]$ is the **mean squared error**, **MSE**: the metric we will use almost always in regression.

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What is the best possible predictor under squared loss? The **conditional mean**:

$$f^*(x) \;=\; \mathbb{E}[Y \mid X = x] \;=\; \arg\min_f R(f)$$

- Intuition: at each x, the constant that minimizes the expected squared error is the average of Y at that x; any other prediction adds the square of its distance to that average (the three-line derivation is in the appendix).

Why regression is the starting point

Regression approximates the conditional mean. That is why it is the course's first method (session 2) and the *benchmark* everything else is measured against.

Reducible and irreducible error

What even the best predictor cannot avoid: for any $\hat{f}$ and any x, with $Y = f(x) + \varepsilon$,

$$\mathbb{E}\big[(Y - \hat{f}(x))^2\big] = \underbrace{\big(f(x) - \hat{f}(x)\big)^2}_{\textbf{reducible}} + \underbrace{\sigma^2}_{\textbf{irreducible}}$$

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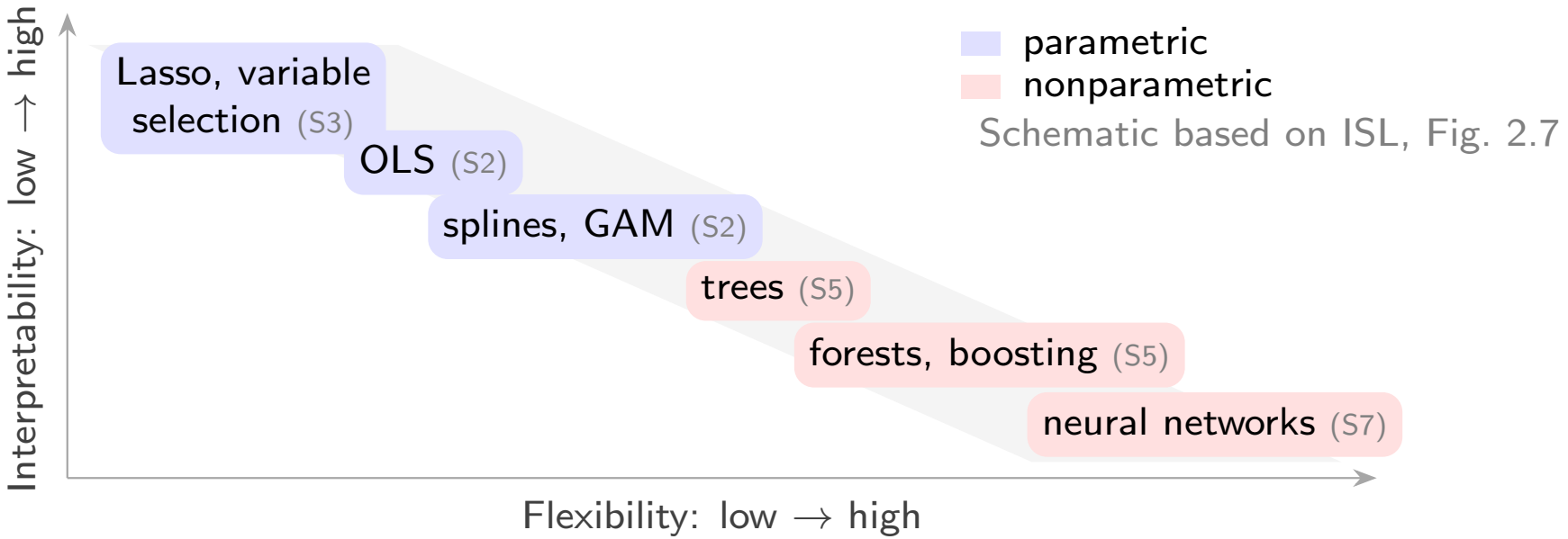
- **Reducible:** how far our $\hat{f}$ is from the true f. This is what a better method, more data, or better variables can improve.
- **Irreducible** (σ^2): not even the perfect model ($\hat{f} = f$) removes it. It is the ceiling on what can be learned with *these* predictors. (The two-line derivation is in the appendix.)

Message

If a model's performance looks "bad", the cause may be a high σ^2: there is not always a better algorithm; sometimes what is missing is better **data**. With only an apartment's area, the ceiling for predicting its price is low and no algorithm raises it: location, stratum, and condition are missing.

Parametric or not, flexible or interpretable

How to estimate f. Parametric: fix the form ($f = \beta_0 + \beta_1 x + \cdots$) and estimate a few numbers; stable, but it can be *misspecified*. **Nonparametric:** let the data draw the shape (k-NN, trees); flexible, but it needs *lots* of data.



The price of flexibility, and why session 6 exists

Moving down the diagonal buys accuracy and loses readability (and risks fitting the noise). Variable importance, partial dependence, and SHAP recover part of the readability.

Empirical risk and its trap

1. The target cannot be computed. $R(f) = \mathbb{E}[L(Y, f(X))]$ averages the loss over *all* possible data: it requires knowing P.

2. What we can compute: the training error. We replace the expectation with the average over our n data points, the **empirical risk**:

$$\hat{R}(f) \;=\; \frac{1}{n}\sum_{i=1}^{n} L\big(y_i, f(x_i)\big)$$

And we give it two jobs: **choosing** the model, $\hat{f} = \arg\min_{f \in \mathcal{F}} \hat{R}(f)$ (*empirical risk minimization*), and **grading** it, by reporting $\hat{R}(\hat{f})$.

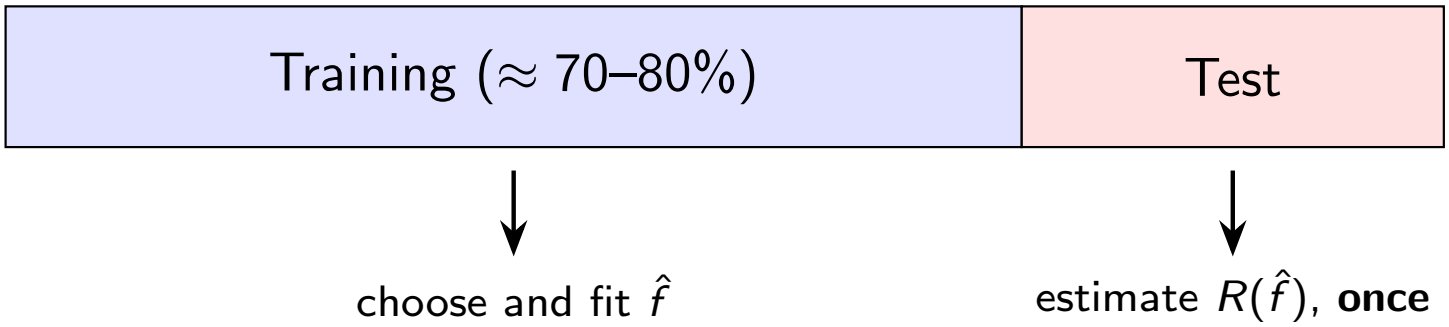
3. The trap: using the same data for both jobs. $\hat{f}$ was chosen *precisely* so that $\hat{R}$ would be small; that is why $\hat{R}(\hat{f})$ is **optimistic** as an estimate of $R(\hat{f})$: it is grading an exam taken with the answer key in hand. In the extreme, a very flexible $\mathcal{F}$ **memorizes** the data ($\hat{R}(\hat{f}) = 0$) without learning anything useful for a new data point.

Golden rule #1

The training error does **not** estimate the generalization error. To grade $\hat{f}$ we need data that played no part in choosing it (next slide).

Training vs. test

The simplest solution: split the data **before** touching any model.



- The **test** set simulates the "new data": its error does estimate $R(\hat{f})$ honestly.
- **Golden rule #2:** the test set is used **at the end, exactly once**. If we use it many times to choose the model, it becomes — in practice — a second training set.
- This is all we need for session 2. What if the sample is small and splitting it "hurts"? That is what **cross-validation** is for (session 3).

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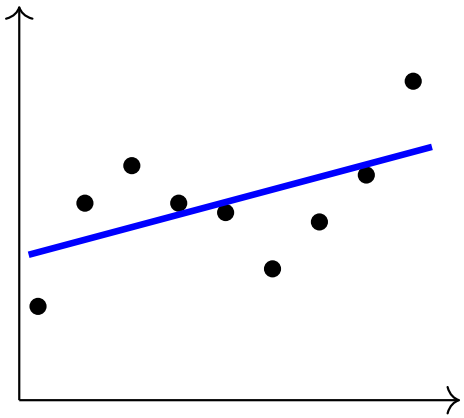
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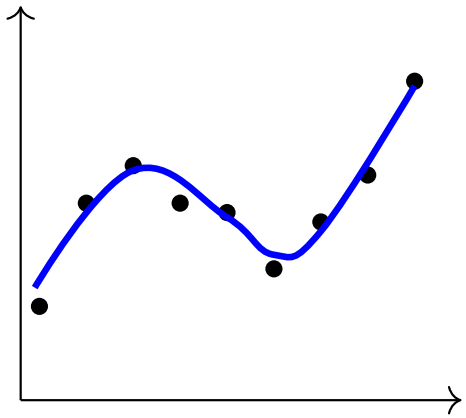
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One dataset, three models

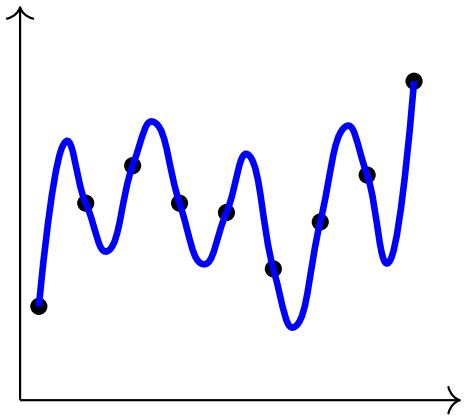
Let us fit polynomials of increasing degree to the same 9 points:



Degree 1: **underfit**



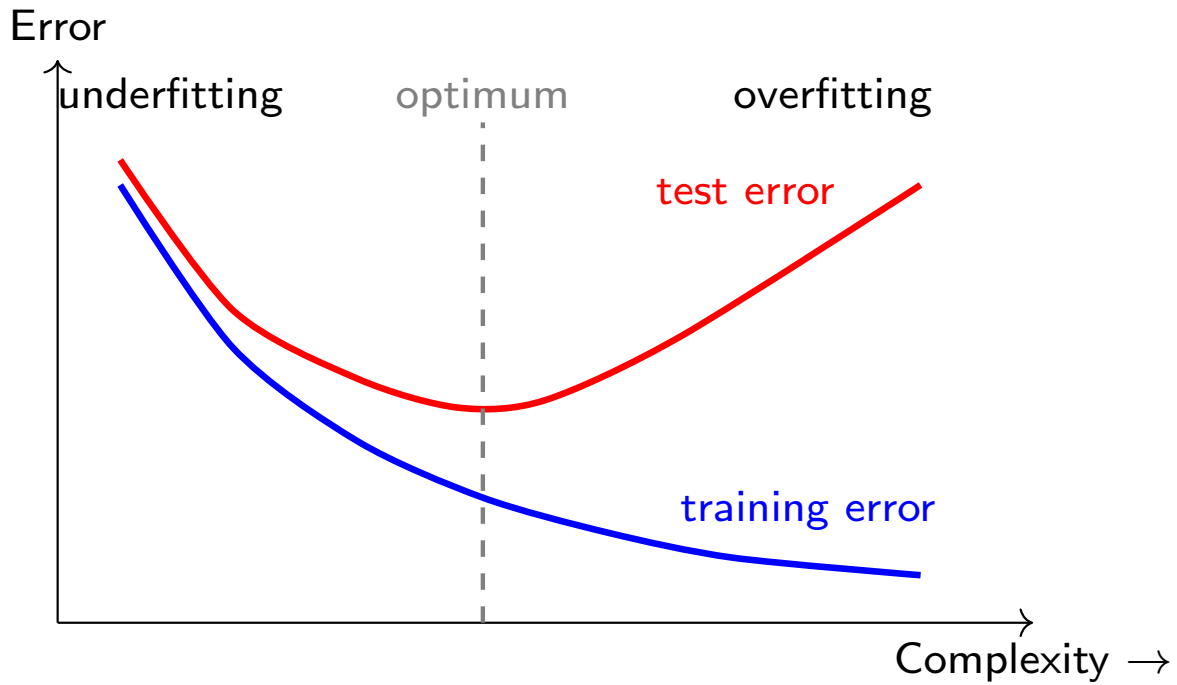
Degree 3: $\approx f$



Degree 15: **overfit**

- **Degree 1:** too rigid — it misses the shape of f (error from **bias**).
- **Degree 15:** passes through *every* point — it fits the **noise** ε, not the signal (error from **variance**).
- Which one best predicts a **new** point? The middle one. Neither the simplest nor the one that "fits best".

The curve that summarizes the course



- The **training** error falls *monotonically* with complexity: more flexibility always fits the sample better.
- The **test** error is **U**-shaped: it improves up to a point and then overfitting destroys it.
- **The whole course lives in this figure:** every method comes with a complexity "knob" and a way to find the minimum of the red curve.

What exactly is "complexity"?

Informally: the **capacity** of the class $\mathcal{F}$ to fit arbitrary patterns.

- In polynomial regression: the **degree** of the polynomial.
- In multiple regression: the **number of predictors** p.
- Later in the course, each method will have its own knob:
  - k-NN: the number of neighbors k (sessions 2 and 4).
  - Regularization: the penalty parameter λ (session 3).
  - Trees: the depth, the leaf size, and the number of trees (session 5).
  - Neural networks: layers, neurons, *dropout* (session 7).

Key idea

Choosing the model is choosing **where to stand** on the U curve. We need: (i) a complexity knob and (ii) an honest estimator of the test error to turn it. Today we have the train/test split; session 3 builds the definitive estimator.

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The thought experiment

We want to understand *why* the test error is U-shaped. Fix a point x_0 and think about what would happen if we could repeat the sampling:

- The training sample $\mathcal{D}$ is **random**: with another n data points, we would get **another** $\hat{f}$.
- Therefore $\hat{f}(x_0)$ is a **random variable**: it changes from sample to sample.
- Two natural objects:
  - $\mathbb{E}[\hat{f}(x_0)]$: the *average* prediction across possible samples.
  - $\mathrm{Var}(\hat{f}(x_0))$: how much the prediction *changes* from sample to sample.
- We define the method's **bias** at x_0:

$$\mathrm{Bias}(x_0) \;=\; \mathbb{E}[\hat{f}(x_0)] - f(x_0)$$

Question

A new data point (x_0, Y_0) arrives, with $Y_0 = f(x_0) + \varepsilon_0$. What is the mean squared error of the prediction at x_0, $\mathrm{MSE}(x_0) = \mathbb{E}[(Y_0 - \hat{f}(x_0))^2]$, averaging over samples $\mathcal{D}$ and over ε_0?

The bias–variance decomposition

Three pieces, and only two are up to us

Result for the mean squared error at x_0, $\mathrm{MSE}(x_0) = \mathbb{E}[(Y_0 - \hat{f}(x_0))^2]$ (appendix):

$$\mathrm{MSE}(x_0) = \underbrace{\left(f(x_0) - \mathbb{E}[\hat{f}(x_0)]\right)^2}_{\textbf{Bias}^2} + \underbrace{\mathrm{Var}\left(\hat{f}(x_0)\right)}_{\textbf{Variance}} + \underbrace{\sigma^2}_{\textbf{Irreducible}}$$

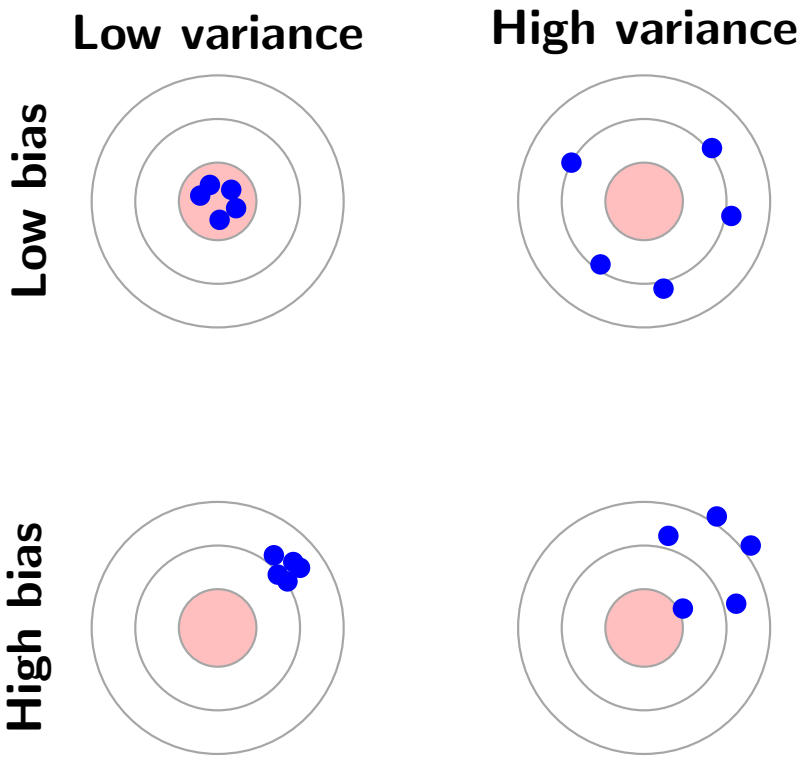
- **Bias:** the method's systematic error. A rigid model (the degree-1 line) errs in the same direction every time, with any sample.
- **Variance:** sensitivity to the particular sample. A flexible model (degree 15) changes completely if a few points change.
- σ^2: noise nobody can remove; the ceiling on what can be learned with these predictors.

Why it matters

Complexity **lowers the bias and raises the variance**. Since the total error is the sum, the optimum is interior: that is the U curve, and every knob in the course (degree, k, λ, depth, epochs) moves along it.

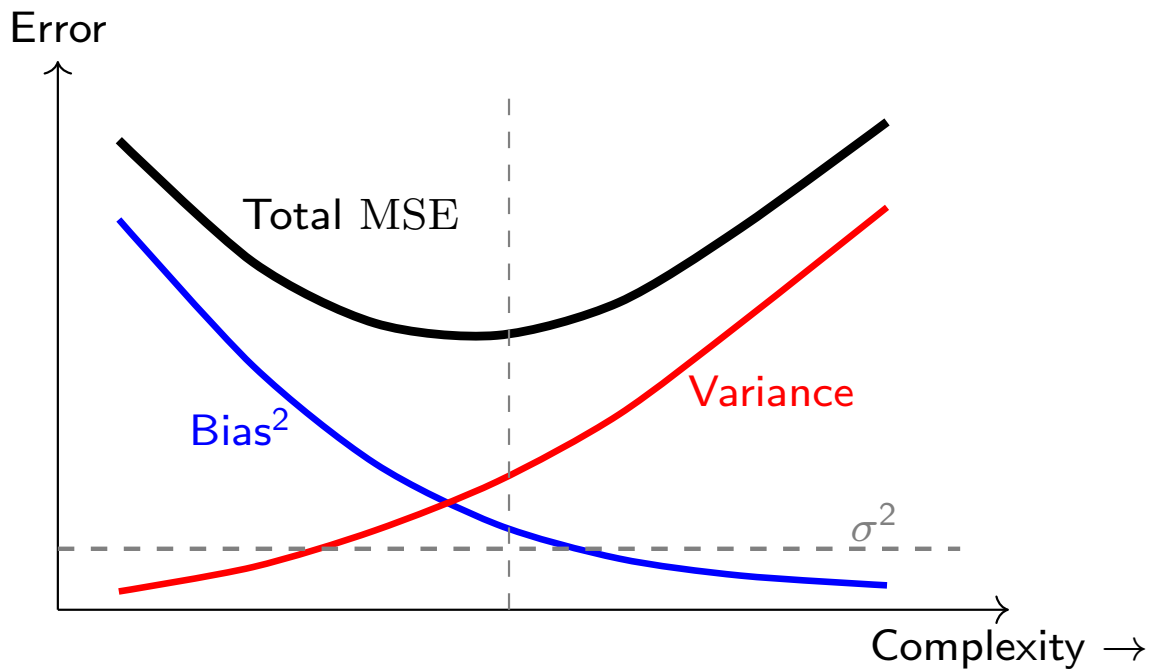
Intuition: the four targets

Each dot is the prediction $\hat{f}(x_0)$ obtained with a different training sample; the center is the true value $f(x_0)$:



- **Bias:** is the *center* of the cloud on the bullseye?
- **Variance:** how *spread out* is the cloud?

The bias–variance trade-off



- More complexity: the bias^2 **falls** (less rigidity) but the variance **rises** (more sensitivity to the sample).
- The total MSE is the **sum** of the three pieces: its minimum is an **interior point** — neither the simplest model nor the most flexible.
- This is the formal explanation of the previous section's U curve.

Why would an economist accept a biased estimator?

The Econometrics I reflex: under Gauss–Markov, OLS is **BLUE** — minimum variance... *among the unbiased*. Unbiasedness is imposed as an **a priori** restriction.

Prediction's logic is different: we minimize the total MSE,

$$\mathrm{MSE} \;=\; \mathrm{Bias}^2 + \mathrm{Variance} + \sigma^2,$$

and nobody gets paid for unbiasedness if the variance explodes.

When does OLS's variance explode? With many predictors relative to n, with highly correlated predictors, with small or noisy samples. In those cases, a **slightly biased** but much **more stable** estimator predicts better.

A teaser for session 3

Ridge and Lasso make exactly that trade, with a knob λ that controls how much bias we buy in exchange for how much variance. (In fact, not even in pure estimation is unbiasedness sacred: James & Stein, 1961.)

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Supervised and unsupervised: the task map

Task	Is there a y?	Question	Example	Session
Regression	Yes (continuous)	how much?	home prices	2–3, 5–7
Classification	Yes (discrete)	which class?	win at the station?	4–5
Clustering	No	which groups?	segment households	7
Dim. reduction	No	what structure?	socioeconomic index	7

- **Supervised:** we know the answer y in the training data; the algorithm learns the map $x \to y$. The "supervision" is the observed y.
- **Unsupervised:** there is no y; we look for structure in the x's (groups, latent factors, dimensions). Without a y there is no test error either: validating is a different game (session 7).
- Images (session 7) combine both worlds: first they are *represented*, then we predict with them.

The two cultures of modeling (Breiman, 2001)

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- **Culture 1 — data modeling:** assume a stochastic model for the box (linear, logit...), estimate its parameters, and validate with assumptions and goodness of fit. In 2001, $\approx 98\%$ of statisticians (and nearly all of econometrics).
- **Culture 2 — algorithmic modeling:** treat the box as **unknown**; look for any $f(x)$ that predicts well and validate with *out-of-sample* error. Back then, $\approx 2\%$; today, the entire industry.

The two cultures of modeling (Breiman, 2001)

Every empirical question starts from the same diagram: nature maps inputs to outcomes.



- **Culture 1 — data modeling:** assume a stochastic model for the box (linear, logit...), estimate its parameters, and validate with assumptions and goodness of fit. In 2001, $\approx 98\%$ of statisticians (and nearly all of econometrics).
- **Culture 2 — algorithmic modeling:** treat the box as **unknown**; look for any $f(x)$ that predicts well and validate with *out-of-sample* error. Back then, $\approx 2\%$; today, the entire industry.

Where does this course stand?

On the border, which has already dissolved: econometrics adopted the algorithms and ML learned causality (a bridge at the close of the course). But we adopt culture 2's **validation discipline**: the test of a model is predicting data it never saw.

Guided application: seeing bias and variance

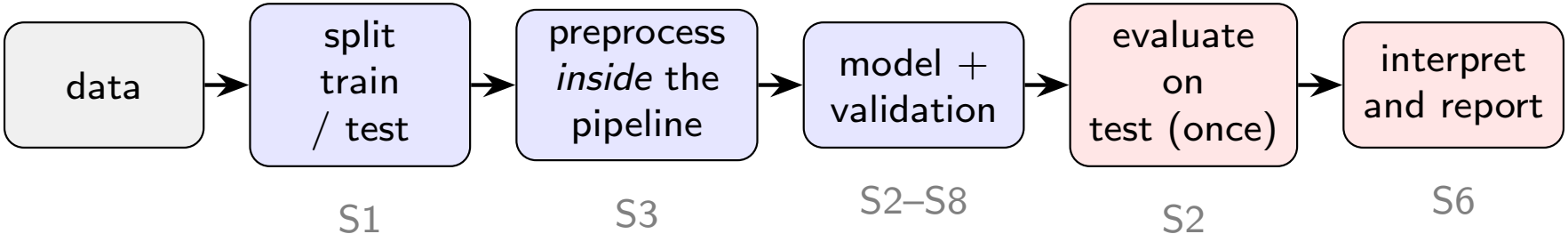
Fifty simulated listings, three polynomials, and one new listing (`applications/week-01`)

1. **The data:** 50 simulated apartment listings: area (m^2) and price (millions of pesos).
2. **Three models on the same 50 points:** polynomials in area of degree 1, 3, and 12. Within the sample the error (root MSE) falls with the degree: 74, 62, and 57. The most flexible one "fits best".
3. **A new listing arrives:** 185 m^2 at 720 million. The line says 811, the cubic 750, degree 12 says 488. We add it and refit: the line and the cubic move by less than 7; the degree-12 curve changes shape (up to 755).
4. **Ten new listings arrive:** the error on them is 69, 56, and 217. The model that fit best within the sample is, by far, the one that predicts worst outside it.
5. **One at a time:** we refit with each of the ten. Ten identical lines, ten nearly identical cubics, ten different degree-12 curves: variance, live.

What to look for

The rigid model errs the same way every time (bias); the flexible one changes with every data point (variance); the one that best predicts a new listing is in the middle.

The course workflow in one slide



Each session changes one piece and holds the rest fixed:

- **S2, S4:** which benchmark model to use and how to measure the error (regression and classification).
- **S3:** how to validate without fooling yourself (CV, *leakage*, spatial and temporal structure).
- **S5, S7:** ever more flexible classes $\mathcal{F}$, each with its own knob.
- **S6:** how to open the box; **S7:** what to do without a y; **S8:** when prediction helps estimate β.

Message

If you understand today's session — risk, overfitting, bias–variance, training and test — you understand *why* every piece of the rest of the course exists.

Recap

1. **Y, not β:** predicting and explaining are different questions. A significant coefficient is not a prediction, and many policy problems are prediction problems.

2. The goal of supervised ML is to minimize the **generalization risk** $R(f) = \mathbb{E}[L(Y, f(X))]$, not the in-sample error; the best predictor under squared loss is the conditional mean.

3. **Golden rule #1:** the training error is optimistic. **Golden rule #2:** the test set is used at the end, exactly once.

4. **Fundamental decomposition:** $\mathrm{MSE} = \mathrm{Bias}^2 + \mathrm{Variance} + \sigma^2$. Complexity lowers the bias and raises the variance: the optimum is interior (the U curve).

5. Moving down the flexibility–interpretability diagonal buys accuracy and loses readability; session 6's tools recover part of the readability.

Getting ready for next class...

To get the most out of next session

This week's required readings:

- ISL, chs. 1 and 2.
  Guide: section 2.2 is today's; figures 2.7 and 2.12 are the ones from class.
- Mullainathan & Spiess (2017), *Machine Learning: An Applied Econometric Approach*, JEP.
  Guide: focus on the $\hat{y}$ vs. $\hat{\beta}$ distinction and on the housing-price example.

Session 2's paper (not required reading; we will make room in class to discuss it): Kim & Zilinsky (2024), *Division Does Not Imply Predictability*, *Political Behavior*. Guide, if you read ahead: Figure 2 and the section "Growing Marginal Effects Do Not Imply Rising Joint Accuracy".

R leveling: complete the introduction and `dplyr` modules (links on the leveling slide) **before** session 2.

Next session: linear regression as a prediction machine (ISL ch. 3 with fresh eyes: what carries over from the OLS you know, what stops mattering, and what is new), k-NN as a first contrast (sec. 3.5), and how to evaluate a prediction: RMSE, MAE, out-of-sample R^2, predicted vs. observed. **R application:** today's 50 listings, with benchmark, line, and test.

Essential references

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- Mullainathan, S., & Spiess, J. (2017). Machine Learning: An Applied Econometric Approach. *JEP*, 31(2), 87–106.
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6 Appendix: going deeper

A1. The conditional mean minimizes the squared risk

Condition on $X = x$ and look for the best constant prediction c:

1. Expand:

$$\mathbb{E}\big[(Y - c)^2 \mid X = x\big] = \mathrm{Var}(Y \mid X = x) + \big(\mathbb{E}[Y \mid X = x] - c\big)^2$$

2. Minimize over c: the first term does not depend on c; the second is ≥ 0 and vanishes at

$$c^* \;=\; \mathbb{E}[Y \mid X = x]$$

3. Conclusion:

$$f^*(x) \;=\; \mathbb{E}[Y \mid X = x] \;=\; \arg\min_f R(f)$$

Interpretation

The **conditional mean** is the theoretical target of the whole course. That is why regression — which approximates it — is ML's natural starting point.

A2. Reducible and irreducible error: the derivation

Take any (fixed) predictor $\hat{f}$ and evaluate at a fixed $X = x$, with $Y = f(x) + \varepsilon$:

1. Substitute and expand:

$$\mathbb{E}\big[(Y-\hat{f}(x))^2\big] = \mathbb{E}\big[(f(x)-\hat{f}(x)+\varepsilon)^2\big] = (f(x)-\hat{f}(x))^2+2(f(x)-\hat{f}(x))\,\mathbb{E}[\varepsilon]+\mathbb{E}[\varepsilon^2]$$

2. Since $\mathbb{E}[\varepsilon] = 0$, the cross term disappears:

$$\mathbb{E}\big[(Y-\hat{f}(x))^2\big] \;=\; \underbrace{\big(f(x)-\hat{f}(x)\big)^2}_{\textbf{reducible}} \;+\; \underbrace{\mathrm{Var}(\varepsilon)}_{\textbf{irreducible}}$$

- **Reducible:** how far our $\hat{f}$ is from the true f. This is what a better method can improve.
- **Irreducible** (σ^2): not even the perfect model ($\hat{f} = f$) removes it. It is the ceiling on what can be learned with these predictors.

A3. Bias–variance derivation (I): separating the irreducible error

Starting point (ε_0 is independent of $\mathcal{D}$, with $\mathbb{E}[\varepsilon_0] = 0$, $\mathrm{Var}(\varepsilon_0) = \sigma^2$):

$$\mathrm{MSE}(x_0) \;=\; \mathbb{E}\big[(Y_0 - \hat{f}(x_0))^2\big], \qquad Y_0 = f(x_0) + \varepsilon_0$$

1. Substitute Y_0 and expand the square:

$$\mathrm{MSE}(x_0) = \mathbb{E}\big[(f(x_0) - \hat{f}(x_0))^2\big] \;+\; 2\,\mathbb{E}\big[(f(x_0) - \hat{f}(x_0))\,\varepsilon_0\big] \;+\; \mathbb{E}[\varepsilon_0^2]$$

2. The cross term vanishes: ε_0 is *new* noise, independent of the sample used to build $\hat{f}$, and has mean zero:

$$\mathbb{E}\big[(f(x_0) - \hat{f}(x_0))\,\varepsilon_0\big] = \mathbb{E}\big[f(x_0) - \hat{f}(x_0)\big] \cdot \underbrace{\mathbb{E}[\varepsilon_0]}_{=0} = 0$$

3. Partial result:

$$\mathrm{MSE}(x_0) \;=\; \mathbb{E}\big[(f(x_0) - \hat{f}(x_0))^2\big] \;+\; \sigma^2$$

The first term is the **estimation error**; we still have to decompose it.

A4. Bias–variance derivation (II): squared bias plus variance

4. Add and subtract $\mathbb{E}[\hat{f}(x_0)]$ inside the estimation error:

$$\mathbb{E}\Big[(f-\hat{f})^2\Big] = \mathbb{E}\Big[\big(\underbrace{f-\mathbb{E}[\hat{f}]}_{\text{constant}} + \underbrace{\mathbb{E}[\hat{f}]-\hat{f}}_{\text{random}}\big)^2\Big]$$

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$$= \big(f-\mathbb{E}[\hat{f}]\big)^2 + \mathbb{E}\Big[\big(\hat{f}-\mathbb{E}[\hat{f}]\big)^2\Big] + 2\big(f-\mathbb{E}[\hat{f}]\big)\underbrace{\mathbb{E}\big[\mathbb{E}[\hat{f}]-\hat{f}\big]}_{=0}$$

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5. Putting it all together:

$$\mathrm{MSE}(x_0) = \underbrace{\big(f(x_0)-\mathbb{E}[\hat{f}(x_0)]\big)^2}_{\textbf{Bias}^2} + \underbrace{\mathrm{Var}\big(\hat{f}(x_0)\big)}_{\textbf{Variance}} + \underbrace{\sigma^2}_{\textbf{Irreducible}}$$

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$$\mathrm{MSE}(x_0) = \underbrace{\big(f(x_0)-\mathbb{E}[\hat{f}(x_0)]\big)^2}_{\text{Bias}^2} + \underbrace{\mathrm{Var}\big(\hat{f}(x_0)\big)}_{\text{Variance}} + \underbrace{\sigma^2}_{\text{Irreducible}}$$

The three pieces of the prediction error

Bias: the method's systematic error (rigidity). **Variance:** sensitivity to the particular sample. σ^2: noise nobody can remove.